

zkBugs Evaluation: Direct vs Original

Common bugs (ran in both modes): **56**.

Verdict transitions FN→TP: **4**, TP→FN: **8**.

Per-tool comparison on the common bug set

Tool	D-TP	D-FN	D-Un	O-TP	O-FN	O-Un	Δ TP	Δ FN
circumspect	0	56	0	1	55	0	+1	-1
circum_civer	3	11	0	1	13	0	-2	+2
picus	3	34	0	0	30	0	-3	-4
ecneproject	13	37	0	7	28	0	-6	-9
zkfuzz	13	16	0	4	12	0	-9	-4
conscs	4	42	0	1	22	0	-3	-20

Cells whose verdict changed between modes

Tool	Direct	Original	Bug
circom_civer	FN	Failure	openzeppelin_mismatched_base64url_decoding_may_break_completeness
circom_civer	FN	Failure	openzeppelin_non_determinism_of_some_inputs
circom_civer	FN	Timeout	daira_hopwood_darkforest_v0_3_missing_bit_length_check
circom_civer	TP	FN	veridise_arrayxor_is_under_constrained
circom_civer	TP	Timeout	koukyosyumei_under_constrained_date_encoder
circom_civer	Timeout	FN	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
circom_civer	Timeout	FN	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
circom_civer	Timeout	FN	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
circom_civer	Timeout	FN	veridise_template.CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
circom_civer	Timeout	Failure	koukyosyumei_unconstrained_base64_decoded_len
circumspect	FN	TP	leastauthority_previously_correct_ownership_proof_disabled_via_code_changes
conscs	FN	Failure	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
conscs	FN	Failure	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
conscs	FN	Failure	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
conscs	FN	Failure	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
conscs	FN	Timeout	matterlabs_emailauth_fails_with_overlapping_invitation_code_regex
conscs	FN	Timeout	matterlabs_fromaddrregex_allows_email_address_spoofing
conscs	FN	Timeout	matterlabs_underconstrained_fpmul_circuit
conscs	FN	Timeout	openzeppelin_mismatched_base64url_decoding_may_break_completeness
conscs	FN	Timeout	veridise_babyjubjub_suborder_constraints_not_applied_correctly
conscs	FN	Timeout	veridise_blacklist_states_not_representable_in_field
conscs	FN	Timeout	veridise_bypassing_internal_transfer_limits_via_swaps
conscs	FN	Timeout	veridise_data_escrow_encrypted_message_wrong_input
conscs	FN	Timeout	veridise_fortxreward_abstracted_away_in_reward_calc
conscs	FN	Timeout	veridise_unsafe_num2bits_254_blacklist_leaf
conscs	FN	Timeout	veridise_zone_id_inclusion_prover_bypass
conscs	FN	Timeout	zksecurity_an_attacker_can_craft_a_fake_non_inclusion_proof_for_a_given_key_due_to_an_aliasing_bug_in_the_smt_verifier
conscs	FN	Timeout	zksecurity_big_integer_zero_check_is_not_sound
conscs	FN	Timeout	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
conscs	FN	Timeout	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrary_input
conscs	FN	Timeout	zksecurity_second_pre_image_attacks_on_packbytesandposeidon_may_be_used_to_register_arbitrary_passports_and_dsc_certificates
conscs	FN	Timeout	zksecurity_the_registration_and_disclosure_circuits_lack_range_checks_for_the_input_indices
conscs	TP	Failure	veridise_arrayxor_is_under_constrained
conscs	TP	Failure	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
conscs	TP	Timeout	koukyosyumei_under_constrained_date_encoder
conscs	Timeout	FN	yacademy_input_signals_is_not_constrained_in_eff_ecdsa_circom
conscs	Timeout	Failure	veridise_template.CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
ecneproject	FN	Failure	matterlabs_email_spoofing_via_manipulated_from_header
ecneproject	FN	Failure	matterlabs_emailauth_fails_with_overlapping_invitation_code_regex
ecneproject	FN	Failure	matterlabs_fromaddrregex_allows_email_address_spoofing
ecneproject	FN	Failure	openzeppelin_non_determinism_of_some_inputs

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ecneproject	FN	Failure	veridise_blacklist_states_not_representable_in_field
ecneproject	FN	Failure	veridise_data_escrow_encrypted_message_wrong_input
ecneproject	FN	Failure	veridise_unsafe_num2bits_254_blacklist_leaf
ecneproject	FN	Failure	veridise_zone_id_inclusion_prover_bypass
ecneproject	FN	Failure	zksecurity_big_integer_zero_check_is_not_sound
ecneproject	FN	Failure	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrar
ecneproject	FN	Failure	zksecurity_sha256_templates_return_zero_on_arbitrary_inputs
ecneproject	FN	Failure	zksecurity_the_registration_and_disclosure_circuits_lack_range_checks_for_the_input_indices
ecneproject	FN	TP	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
ecneproject	FN	TP	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
ecneproject	FN	TP	zksecurity_second_pre_image_attacks_on_packbytesandposeidon_may_be_used_to_register_arbitrary_passports_and_dsc_certificates
ecneproject	FN	Timeout	openzeppelin_mismatched_base64url_decoding_may_break_completeness
ecneproject	Failure	FN	veridise_zone_limits_bypass_zswapv1
ecneproject	TP	FN	koukyosyumei_under_constrained_date_encoder
ecneproject	TP	FN	matterlabs_underconstrained_fpmul_circuit
ecneproject	TP	FN	yacademy_input_signal_s_is_not_constrained_in_eff_ecdsa_circom
ecneproject	TP	FN	yacademy_under_constrained_circuits_compromising_the_soundness_of_the_system
ecneproject	TP	Failure	koukyosyumei_unconstrained_base64_decoded_len
ecneproject	TP	Failure	veridise_arrayxor_is_under_constrained
ecneproject	TP	Failure	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
ecneproject	TP	Failure	zksecurity_an_attacker_can_craft_a_fake_non_inclusion_proof_for_a_given_key_due_to_an_aliasing_bug_in_the_smt_verifier
ecneproject	TP	Timeout	veridise_template_CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
ecneproject	Timeout	FN	veridise_kyt_signature_verification_fails_nonzero_hash
ecneproject	Timeout	FN	veridise_nullifier_verification_can_be_disabled
ecneproject	Timeout	Unknown	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
picus	FN	Failure	openzeppelin_mismatched_base64url_decoding_may_break_completeness
picus	FN	Failure	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
picus	FN	Failure	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
picus	FN	Failure	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
picus	FN	Failure	zksecurity_second_pre_image_attacks_on_packbytesandposeidon_may_be_used_to_register_arbitrary_passports_and_dsc_certificates
picus	FN	Failure	zksecurity_the_registration_and_disclosure_circuits_lack_range_checks_for_the_input_indices
picus	FN	Timeout	openzeppelin_non_determinism_of_some_inputs
picus	FN	Timeout	zksecurity_big_integer_zero_check_is_not_sound
picus	Failure	Timeout	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrar
picus	TP	FN	koukyosyumei_under_constrained_date_encoder
picus	TP	Failure	koukyosyumei_subtract_unconstrained_multiplier
picus	TP	Failure	veridise_arrayxor_is_under_constrained
picus	Timeout	FN	matterlabs_underconstrained_fpmul_circuit
picus	Timeout	FN	yacademy_input_signal_s_is_not_constrained_in_eff_ecdsa_circom
picus	Timeout	FN	yacademy_under_constrained_circuits_compromising_the_soundness_of_the_system
picus	Timeout	Failure	koukyosyumei_extractstringfrompoint_shiftedfirstbyte_unconstrained

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picus	Timeout	Failure	koukyosyumei_unconstrained_base64_decoded_len
picus	Timeout	Failure	matterlabs_emailauth_fails_with_overlapping_invitation_code_regex
picus	Timeout	Failure	matterlabs_fromaddrregex_allows_email_address_spoofing
picus	Timeout	Failure	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
picus	Timeout	Failure	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
picus	Timeout	Failure	veridise_missing_range_checks_in_bigmod
picus	Timeout	Failure	veridise_template_CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
picus	Timeout	Failure	veridise_underconstrained_circuit_allows_invalid_comparison
picus	Timeout	Failure	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
picus	Timeout	Failure	zksecurity_missing_byte_range_checks_allows_packed_data_pollution
zkfuzz	FN	Failure	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
zkfuzz	FN	Failure	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
zkfuzz	FN	Failure	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
zkfuzz	FN	Failure	zksecurity_big_integer_zero_check_is_not_sound
zkfuzz	FN	Timeout	daira_hopwood_darkforest_v0_3_missing_bit_length_check
zkfuzz	FN	Timeout	matterlabs_underconstrained_fpmul_circuit
zkfuzz	FN	Timeout	veridise_incorrect_initialization_in_membership_circuits
zkfuzz	FN	Timeout	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
zkfuzz	FN	Timeout	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
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zkfuzz	Timeout	Failure	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrary_input
zkfuzz	Timeout	Failure	zksecurity_sha256_templates_return_zero_on_arbitrary_inputs
zkfuzz	Timeout	TP	veridise_missing_range_checks_on_comparison_circuits